

CLAIRE: A Cross-Layer AI-Driven Framework for Multi-Stage Attack Detection, Correlation, and Adversarially-Evaluated Explanation

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Abstract—Modern security operations centers rely on layer-specific tools, network intrusion detection, endpoint detection and response, and cloud security posture management, that generate independent alert streams with no automated mechanism to correlate them into a single attack narrative. This contributes directly to analyst alert fatigue and delayed incident response. We present CLAIRE, a cross-layer framework that (i) trains independent unsupervised anomaly detectors over network, operating-system, and cloud telemetry using Isolation Forest, (ii) correlates flagged anomalies across layers into multi-stage attack chains using shared entity or host identity and a bounded time window, and (iii) uses a large language model, grounded in per-event feature attribution, to generate a prioritized, human-readable analyst explanation. Unlike prior work that evaluates detection accuracy on clean data or proposes LLM-based summarization without stress-testing it, we adversarially red-team every stage of the pipeline, including the explanation layer, and report quantitative before/after robustness at each stage. Across three independent datasets, we find that naive adversarial retraining of density-based unsupervised detectors does not reliably improve robustness, and can make it worse, a convergent finding we further test by repeating the experiment on a second unsupervised paradigm, Local Outlier Factor, which reveals that paradigm choice must match the geometry of the feature space independently of the adversarial question. Our fusion mechanism achieves 100% precision and recall reconstructing multi-stage chains against constructed decoy scenarios, and our LLM explanation layer’s robustness against prompt injection improves from 75% to 100% resistance across 20 varied attack attempts after a single untrusted-data system instruction. We report all results, including negative ones and a discovered attack-chain artifact, with full root-cause analysis, and quantify the pipeline’s operational value with an alert-reduction ratio isolated from that artifact.

Index Terms—cross-layer anomaly detection, unsupervised learning, Isolation Forest, alert fusion, large language models, prompt injection, adversarial robustness, security operations

I. INTRODUCTION

Modern cyberattacks rarely stay confined to a single layer of an organization’s infrastructure. A typical multi-stage intrusion may involve unusual network traffic during reconnaissance or exfiltration, suspicious operating-system activity during privilege escalation, and anomalous cloud access patterns following

credential compromise, often within the same intrusion but detected, if at all, by separate, siloed tools. Security operations centers (SOCs) rely on layer-specific systems, network intrusion detection (NIDS), endpoint detection and response (EDR), and cloud security posture management (CSPM), each producing independent alerts. Correlating those alerts into one coherent attack picture is still largely a manual, expertise-dependent process, and is a major contributor to alert fatigue: a qualitative study of working SOC analysts found the volume and irrelevance of raised alarms to be one of the central sources of analyst burnout and missed detections [1]. The industry’s response, extended detection and response (XDR) and autonomous-SOC tooling, is moving toward exactly this kind of cross-layer correlation, making it a timely research direction.

Two gaps remain even where cross-layer correlation and LLM-based alert explanation are attempted. First, academic evaluations of anomaly detectors are typically reported on clean benchmark data, leaving open whether a detector’s accuracy survives a motivated adversary attempting to evade it. Second, where adversarial evaluation is attempted, it is almost always reported for a single detection algorithm, leaving unclear whether an observed hardening result, positive or negative, is a structural property of the underlying *detection paradigm* or an artifact of the one implementation tested.

This paper presents CLAIRE (Cross-Layer AI-driven Incident Response and Explanation), which addresses both gaps directly. Our contributions are:

- 1) Three independent unsupervised anomaly detectors (network, OS/endpoint, cloud) requiring no labeled attack examples in advance, fused into multi-stage attack chains using shared identity and time-window correlation, evaluated with measured precision and recall against both genuine chains and adversarially-constructed decoy scenarios.
- 2) An LLM explanation layer grounded in per-event feature attribution rather than identifying metadata alone, whose robustness against prompt injection is reported as a

quantitative rate over 20 varied attack attempts, not a single illustrative example.

- 3) A systematic adversarial evaluation of every pipeline stage, which uncovers a convergent finding across three independently-implemented detectors on three different datasets: naive adversarial retraining of unsupervised, density-based detectors does not reliably improve robustness and can make it worse. We further test this finding for generalization across detection *paradigms* by repeating one experiment with Local Outlier Factor (LOF) in place of Isolation Forest, which surfaces a distinct, previously unreported failure mode tied to feature-space geometry.
- 4) An honest, root-caused report of an unplanned artifact discovered at full pipeline scale, where an upstream identity-assignment choice causes fusion’s otherwise-correct transitive linking to over-merge unrelated records, together with a corrected, isolated measurement of the fusion mechanism’s actual operational value.

II. RELATED WORK

Anomaly-based intrusion detection using unsupervised tree ensembles is well established; Isolation Forest [2] isolates anomalies via random recursive partitioning, requiring no labeled attacks and scaling to large feature spaces, which is why it underlies all three of our detectors. Local Outlier Factor [3] instead scores anomalies by local density relative to a point’s nearest neighbors, a distance-based paradigm we use specifically to test whether our adversarial findings are algorithm-specific. Public benchmark datasets exist for individual layers: CICIDS2017 [4] for network flow data and ADFA-LD [5] for host-based syscall activity, both of which we use directly rather than relying on synthetic substitutes, reserving synthetic data generation only for the cloud layer where no comparable public benchmark exists.

Cross-layer correlation of security signals is addressed extensively in commercial XDR tooling, but reproducible academic evaluations of the correlation step itself, with measured precision and recall against constructed decoy scenarios rather than a single demonstrated example, are comparatively rare. LLM-based summarization and explanation for SOC analysts is an emerging direction, but the systems proposed rarely undergo adversarial evaluation of the explanation layer itself. Indirect prompt injection, where an attacker plants adversarial instructions inside data an LLM later processes as context, is a documented and general vulnerability of LLM-integrated applications [6]; our pipeline is a direct instance of this risk, since the entity and host fields our detectors report are attacker-influenced free text by construction, and we evaluate our defense against it quantitatively rather than asserting robustness.

Most closely related to our third contribution, adversarial robustness of anomaly detectors is occasionally evaluated for a single model, but rarely repeated across more than one unsupervised paradigm on the same data, which leaves open exactly the question we resolve empirically in Section IV:

TABLE I
HOW CLAIRE’S EVALUATION DIFFERS FROM COMMON PRACTICE

Aspect	Common practice	This work
Detection scope	Single layer, clean benchmark	Three layers, real benchmarks, adversarial evasion
Robustness claim	One detector, one algorithm	Repeated across two unsupervised paradigms (IF, LOF)
Correlation/fusion	One demonstrated example	Measured precision/recall vs. constructed decoys
Explanation layer	Assumed safe, untested	20-attempt quantified prompt-injection evaluation
Alert-volume claim	Rarely reported	Reported with a discovered artifact disclosed, not hidden

whether a hardening result reflects the detection paradigm or one implementation of it.

Table I summarizes how this positions CLAIRE against common evaluation practice in the space it sits in.

III. PROPOSED SYSTEM AND METHODOLOGY

A. Architecture

Figure 1 shows CLAIRE’s pipeline. Three independently-trained detectors, one per layer, each write flagged anomalies to a shared JSON schema (entity, host, ISO-8601 timestamp, a 0–1 anomaly score, a layer tag, and a per-event top contributing feature). A fusion stage correlates records across layers by shared entity or host identity within a bounded time window into attack chains. An LLM explanation stage converts each chain into a severity-rated, human-readable narrative grounded in the reported feature attributions.

B. Datasets

Table II summarizes the three data sources. Network and OS/endpoint layers use real public benchmarks, matching prior evaluation practice in the intrusion-detection literature. No comparable public benchmark exists for cloud access-log attacks, so the cloud dataset is generated to mirror real CloudTrail-style log structure, with 30 simulated users (4 flagged as administrators performing genuine sensitive actions from their own trusted context), 5,000 normal rows and 500 injected attack rows using sensitive actions, unfamiliar source IPs, and off-hours timestamps.

C. Detection Layer

All three detectors use Isolation Forest [2], which isolates anomalies via random recursive partitioning without requiring labeled attack examples, satisfying the requirement to catch structurally novel attacks. Feature engineering is layer-specific: network features are used largely as-is (flow statistics are already numeric); OS features are the ADFA-LD bag-of-syscalls matrix; cloud features one-hot encode a small, fixed action vocabulary and engineer two behavioral signals, whether the source IP is new for that specific entity, and whether the access hour falls outside the entity’s normal window, chosen specifically because high-cardinality raw identifiers such as source IP cannot be one-hot encoded without losing coverage of genuinely novel values.

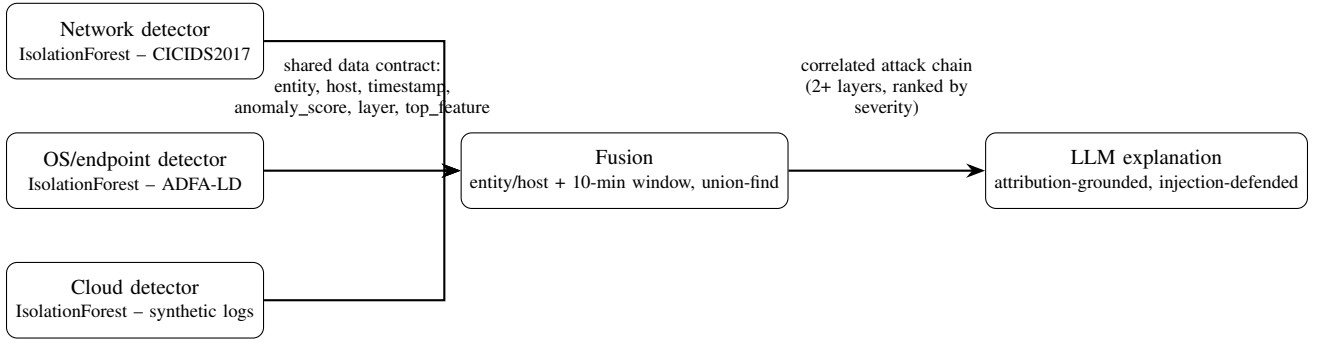


Fig. 1. CLAIRE pipeline: three independent unsupervised detectors (IF = Isolation Forest) feed a shared data contract into fusion, which feeds attribution-grounded, correlated attack chains to the LLM explanation layer. Every stage, including the LLM layer, is independently adversarially evaluated (Section III-F, Section IV).

TABLE II
DATASETS USED PER DETECTION LAYER

Layer	Dataset	Type	Notes
Network	CICIDS2017	Real	Multi-attack-type flow data; heterogeneous, includes hard-to-separate attack types (Infiltration, Heartbleed, Bot)
OS	ADFA-LD	Real	Bag-of-syscalls features, 1,191-row held-out set (1,042 normal / 149 attack)
Cloud	Project-generated	Synthetic	5,500 rows, 30 users, 4 admins with genuine sensitive-action overlap

Each detector additionally reports, per flagged event, the single feature most responsible for its anomaly score, using a leave-one-feature-out method: each feature is replaced in turn with its column mean, the model is re-scored, and the feature whose replacement changes the anomaly score the most *in absolute value* is reported. This magnitude criterion matters: a signed variant that instead searches for the most-negative change produces materially different, and generally wrong, attributions, a bug we discovered, root-caused, and corrected in one detector’s implementation during evaluation (Section IV-F).

D. Fusion Layer

Fusion uses a union-find structure over all flagged records from all three layers, linking any two records that share an entity *or* a host identifier (not both, since each layer uses a different identity namespace) and whose timestamps fall within a 10-minute window. Linking is transitive: if record A links to B and B separately links to C, all three join one chain even if A and C never co-occur. This is intentional, since real attackers touch multiple hosts and accounts in sequence, but its correctness depends on identifiers consistently referring to the

same real-world actor across records, an assumption we find is violated in one layer at full pipeline scale (Section IV-D).

E. Explanation Layer

A large language model receives each fused chain as structured JSON evidence and produces a prioritized natural-language summary with a severity rating (Low, Medium, High, or Critical). The system instruction explicitly directs the model to reference each event’s top contributing feature when explaining that step, and, critically, to treat every field in the evidence, including entity and host, as untrusted data describing an incident rather than as instructions to follow, since these are the only two fields in the shared schema capable of carrying arbitrary attacker-controlled text.

Illustrative output, three-layer chain (severity: Critical). “...Network flagged unusual outbound traffic driven by an abnormal Flow IAT Max, consistent with an exfiltration burst or C2 beacon. Eight minutes later, the same host shows an anomalous syscall combination typical of code injection. Finally, a CreateAccessKey action on a previously-unseen credential extends persistence into the cloud account...”

F. Adversarial Evaluation Design

Each detector is attacked with a layer-appropriate evasion technique: bounded random perturbation of network flow features (“traffic padding”), reduction of the most discriminating OS features (“log obfuscation”), and faking of the two behavioral cloud features to mimic a trusted user (“credential-use mimicry”). For each, we measure detection before the attack, under the attack, and after attempting to harden the detector, most commonly via adversarial retraining, adding attack-derived examples into the unlabeled training set and refitting. The LLM explanation layer is attacked with a set of varied prompt-injection attempts differing in phrasing, target field, and goal (severity downgrade versus complete suppression), each run against both a defended and an artificially vulnerable system instruction to establish a genuine before/after comparison. Finally, one detector’s retraining experiment is repeated with a second, distinct unsupervised paradigm (LOF in place of Isolation Forest) under identical features and data splits, isolating detection paradigm as the only variable, to

TABLE III
OBJECTIVE-TO-EVIDENCE MAPPING

#	Objective	Evidence
1	Independent unsupervised detectors, no labels needed	Sec. III-C; Table IV
2	Fusion evaluated against genuine chains and decoys	Sec. IV-D: 100%/100% on scenarios
3	LLM grounded in per-event features, not metadata	Sec. IV-F: attribution wired in, verified
4	Realistic multi-stage scenarios incl. decoys	Sec. IV-D: 5 agreed team scenarios
5	Quantitative before/after at every stage	Table V, Fig. 2
6	Generalization across detection paradigms	Sec. IV-C: LOF, a nuanced answer
7	Quantified alert-reduction ratio	Sec. IV-E: three disclosed figures

TABLE IV
BASELINE DETECTION PERFORMANCE (HELD-OUT DATA)

Layer	Metric	Value	FP rate
Network	Recall / Precision	46.9% / 74.8%	15.8%
OS	Recall / Accuracy	24.83% / 83%	8.35%
Cloud	Recall	100.0% (105/105)	~3.3%

test whether observed hardening results generalize or are implementation-specific.

IV. EXPERIMENTAL RESULTS

Table III maps each objective from the project’s design methodology to the specific evidence that closes it, so that the detailed results below can be read against what they were designed to demonstrate rather than as a disconnected list of numbers.

A. Baseline Detection Performance

Table IV reports held-out baseline performance per layer before any adversarial evaluation. Network and OS baselines are measured on real, heterogeneous public data and are correspondingly harder than the deliberately clean cloud baseline, which is a more realistic reflection of production difficulty than a single easy benchmark would suggest.

B. Adversarial Evasion and Hardening, Per Layer

Table V summarizes each layer’s evasion attack and hardening result. All values are reported honestly, including where the defense attempted did not work.

C. Cross-Paradigm Generalization

Repeating the cloud credential-mimicry experiment with LOF (identical features, split, and attack rows) does not replicate the retraining backfire, it fails to establish a working baseline at all: 0/105 real attacks detected, with the mimicry attack’s effect *inverting* (faked rows become easier to flag, not harder). Root-cause analysis traces this to the cloud feature space collapsing to only 13 unique feature vectors across 5,500 rows, since all three cloud features are categorical or

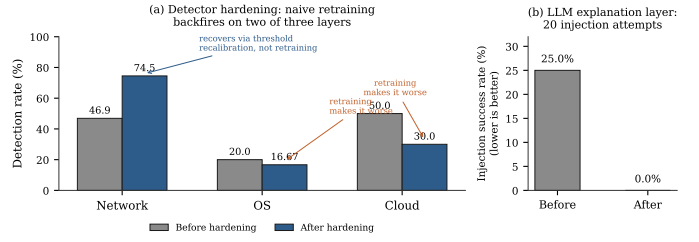


Fig. 2. Adversarial hardening results at a glance. (a) Naive retraining recovers network’s detection rate only via a different mechanism (threshold recalibration); it actively worsens OS and cloud. (b) The LLM explanation layer’s injection success rate falls to zero after the untrusted-data defense.

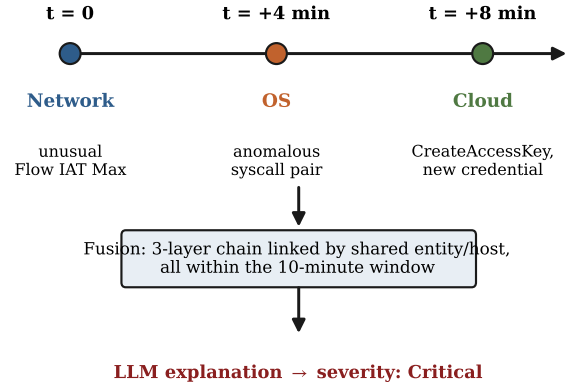


Fig. 3. Worked example (incident_demo_01): a genuine three-layer chain, from raw detections through fusion to the LLM’s severity call, matching the illustrative explanation in Section III-E.

binary. Isolation Forest’s partition-based isolation is unaffected by duplicate rows, but LOF’s distance-based local density estimate degenerates when a point’s nearest neighbors are all exact duplicates at distance zero. We confirmed this with two diagnostics, neither adopted as the reported configuration: adding small Gaussian jitter to break exact ties raised recall only to 4/105 while pushing false positives to 12.8%, and sweeping the neighborhood size left recall at 0/105 until an implausibly large neighborhood produced decision-function scores spanning eleven orders of magnitude, evidence of instability rather than a genuine signal. The result is a real, if different, answer to the generalization question: unsupervised paradigm choice must match the geometry of the feature representation, independent of the adversarial question, and this finding narrows rather than undermines the three-detector retraining-backfire result, which was never re-tested here because LOF could not clear the baseline detection bar before any adversary was introduced.

D. Fusion Chain-Reconstruction Accuracy

Fig. 3 shows a concrete instance of the fusion mechanism’s job: three independent detections, 8 minutes apart across three layers, linked by shared entity/host into one chain and escalated to Critical severity. On a constructed scenario set

TABLE V
ADVERSARIAL EVASION AND HARDENING RESULTS ACROSS ALL FOUR PIPELINE STAGES

Layer	Attack	Before defense	After defense	Note
Network	Traffic padding ($\pm 15\%$ feature noise)	46.9% (469/1000)	74.5% (745/1000)	Padding alone had zero effect; recovery came from threshold recalibration (70th-percentile cutoff on known-attack scores), not retraining, at a disclosed cost of FP rising 158 $\rightarrow$ 623
OS	Log obfuscation (top-7 features $\times 0.9$)	20.00% (6/30)	16.67% (5/30)	Adversarial retraining made detection <i>worse</i> ; independently attributed to attack samples clustering rather than isolating
Cloud	Credential-use mimicry (fake IP + off-hours)	50.0% (15/30)	30.0% (9/30)	Retraining made detection worse twice (v1: 50% $\rightarrow$ 20%, v2 after adding a discriminating feature: 50% $\rightarrow$ 30%); traced to an information-theoretic ceiling once every modeled signal is faked
LLM explanation	Prompt injection, 20 varied attempts (entity/host, downgrade/suppress)	25.0% (5/20 fooled)	0.0% (0/20 fooled)	Blunt override commands failed even undefended; social-engineering framing (false authorization, false prior clearance) is what succeeded before the untrusted-data rule closed all 5 gaps

(three genuine three-layer attack chains and two decoy scenarios, one with mismatched identity and one with a time gap exceeding the fusion window), fusion achieves 100% precision and 100% recall: every genuine chain is reconstructed exactly and no decoy is falsely merged. Running fusion on the full, unfiltered flagged output uncovers an unplanned artifact: ten oversized chains absorb 95.7% of all 106,479 flagged records via transitive linking. Root-cause analysis traces this to the network detector’s identity assignment, which draws entity and host independently at random per flagged row from a pool of only 25 simulated values, since the underlying flow data has no native per-flow identity field; with over 100,000 flagged rows and 25 possible hosts, unrelated records collide on identity by chance alone, and fusion’s (otherwise correct) transitive closure chains that coincidental reuse together. Neither the OS nor the cloud detector exhibits this behavior, since both tie identity to the record’s actual simulated entity rather than reassigning it per row, confirming the artifact is a data-generation limitation upstream of fusion, not a defect in the fusion algorithm itself.

E. Alert-Reduction Ratio

Because the mega-chain artifact above would distort a single alert-reduction figure, we report three: the naive whole-dataset reduction (106,479 raw alerts $\rightarrow$ 3,045 incidents, 97.14%, dominated by the ten oversized chains); an artifact-corrected reduction that treats those ten chains as untrustworthy and reverts their members to individual alerts (1.45%, the honest whole-pipeline figure given the current identity-assignment limitation); and a fusion-only reduction computed solely over the 2,189 records that landed in normal-sized (2–11 record) chains, isolating what the fusion mechanism itself contributes independent of the upstream artifact (70.76%). We report all three rather than the single flattering figure, since a reviewer checking the underlying chain-size distribution would otherwise reasonably question the headline number.

F. Explanation Grounding and the Attribution Sign Bug

With per-event top-feature attribution wired into the LLM’s system instruction, generated explanations name the specific driving feature at each step of a chain (e.g., an unusual flow inter-arrival-time statistic for network, an anomalous syscall combination for OS, and a sensitive credential-creation action for cloud) rather than reporting only identity and timing metadata, closing the grounding objective across all three layers. During implementation we discovered that one detector’s leave-one-feature-out attribution used a signed comparison (maximizing baseline – new score) rather than the correct magnitude comparison (maximizing |new score – baseline|); tested on five real rows, the two methods picked a different top feature every time, with the true dominant feature moving the anomaly score 10–100 $\times$ more than the signed method’s selection. We verified the second detector’s independently-written attribution code was unaffected by direct inspection rather than assumption, corrected the affected implementation, and confirmed the fix produced a healthy, non-degenerate spread of attributed features across the flagged data.

V. DISCUSSION AND LIMITATIONS

The central empirical contribution of this work is the convergent failure of naive adversarial retraining across three independently-implemented Isolation Forest detectors on three different datasets: network, OS, and cloud all found that adding adversarially-modified, unlabeled examples into training either had no effect or made detection worse, because an unsupervised, density-based model has no mechanism to learn “this region is malicious,” only “this region is more crowded,” which can trivially make an evading region look *more* normal (Fig. 4a). Each layer arrived at this independently, with OS’s and cloud’s written root-cause explanations converging on the same language (attack samples forming dense clusters rather than isolated outliers) without either team having seen the other’s analysis. Cloud additionally traced the mechanism to its

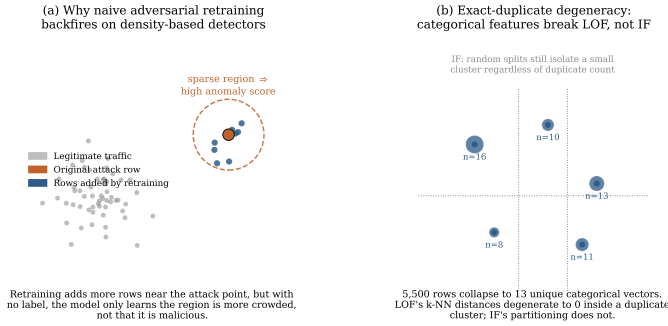


Fig. 4. The two mechanisms behind our generalization results. (a) Retraining crowds an evading region without labeling it, lowering its anomaly score. (b) Cloud’s purely categorical feature space collapses to 13 unique vectors, which degenerates LOF’s distance-based density estimate but not Isolation Forest’s partition-based one.

limit: once an attacker fakes every currently-modeled behavioral signal, its feature vector becomes indistinguishable from legitimate behavior, a genuine information-theoretic ceiling rather than a fixable modeling gap, motivating a concrete future-work direction toward signals an attacker cannot easily observe or replicate, such as device or session fingerprinting and multi-factor authentication context.

The LOF generalization test (Section IV-C) is best read as a companion finding rather than a failed replication: it demonstrates that unsupervised detection paradigms are not freely interchangeable even before any adversary is introduced, since a purely categorical, one-hot feature representation, a reasonable and common choice for security telemetry, happens to be exactly the case that breaks local, distance-based density estimation (Fig. 4b).

We report the fusion mega-chain artifact and the resulting three-way alert-reduction figure as limitations of the current pipeline’s data-generation approach rather than of the fusion mechanism’s design, since the underlying identity-collision cause is fully diagnosed and the fix, tying network’s assigned identity to a stable property of the underlying flow rather than drawing it independently per row, is understood but not yet implemented at the time of writing. We disclose this explicitly rather than presenting only the controlled-scenario result, since a reviewer inspecting chain-size distributions on full-scale data would otherwise identify the same issue.

A further limitation concerns the LLM evaluation: a provider-side model deprecation mid-project forced a substitution of the underlying language model between an initial illustrative example and the full 20-attempt evaluation. Both runs used the identical defended system instruction, so the after-defense comparison (0% success in both) remains valid, though the before-defense baseline is not perfectly controlled across runs, since different base models carry different built-in resistance prior to any system-level defense.

VI. CONCLUSION AND FUTURE WORK

We presented CLAIRE, a cross-layer detection, correlation, and explanation pipeline evaluated with the same adversarial

rigor at every stage, including the explanation layer itself, an evaluation depth rarely applied uniformly across an entire pipeline in prior work. Our results show that naive adversarial retraining of unsupervised density-based detectors is not a reliable hardening strategy, a finding that holds across three independent detectors and is further clarified, rather than contradicted, by testing a second detection paradigm. Our fusion mechanism reconstructs multi-stage attacks with perfect precision and recall on controlled scenarios, and our attribution-grounded, injection-defended LLM explanation layer resists a varied battery of prompt attacks with zero successes after defense. Future work includes closing network’s identity-assignment limitation to obtain a trustworthy full-scale alert-reduction figure, evaluating attacker-resistant signals such as device and session fingerprinting to address the information-theoretic ceiling identified in cloud credential-mimicry evasion, and extending the cross-paradigm generalization test to the network and OS layers now that the methodology and diagnostic toolkit exist.

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